

Dynamic Arrays

Creating a matrix “package”

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- Lets write the `multiplyMatrices` function.
Remember matrix multiplication is defined as follows:

For matrices A of dimension $m \times n$ and B of dimension $n \times p$:

$$C_{ij} = \sum_{k=0}^{n-1} A_{ik} \times B_{kj}, 0 \leq i < m, 0 \leq j < p$$

where C is the resulting matrix of dimension $m \times p$.

- The following is an example of a function call:

```
float** multiplyResult = multiplyMatrices(matrixA,2,3,matrixC,3,4);
```

Provide the function definition.

Write an implementation for the function.

- Now create a library of matrix functions. An incomplete header file is given by:

```
#ifndef MATRIX_H
#define MATRIX_H

float** allocateMatrix(int, int);
void deallocateMatrix(float**, int);

void readMatrix(float**, int, int);
void printMatrix(float**, int, int);

float** addMatrices(float**, float**, int, int);
float** multiplyMatrices(float**, int, int, float**, int, int);

#endif
```

- Add all implementations in the .cpp file for the functions in the header file. Some have been given in previous slides.
- Write a main function to test your functions.

- Create a dynamic array library. The header file for the library is given by:

```
#ifndef DynamicArray_h
#define DynamicArray_h

int getSize();
void newArray(int);
void initialiseArray();
void putValue(int, int);
int getValue(int);
void resize(int);
void deleteArray();

#endif /* DynamicArray_h */
```

- Questions to ask!
 - Where is the dynamic array declared and initialised? Where is the size declared and defined? Is this the best solution?
 - What functionality do each of the given functions provide?
 - Provide a main function to make use of the DynamicArray functionality as it currently stands.